



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 12

Subject:	DevOps Lab
TCODE:	TE7950
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Branch	CSE, Batch (2023-27), L4
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23 rd July, 2026
Title of Assignment:	Kubernetes Pod & Deployment
Theory:	Kubernetes is an open-source container orchestration platform used to automate the deployment, scaling, and management of containerized applications. A Pod is the smallest deployable unit in Kubernetes and contains one or more containers that share the same network and storage resources. A Deployment manages Pods by maintaining the desired number of replicas, automatically replacing failed Pods, and supporting updates without downtime. Kubernetes simplifies application management by ensuring high availability, scalability, and efficient resource utilization across a cluster. Commands Used • Checked the Kubernetes client version: kubectl version --client • Verified the Kubernetes cluster information: kubectl cluster-info • Displayed the available cluster nodes: kubectl get nodes • Created the Kubernetes Deployment from the YAML manifest: kubectl apply -f web-deployment.yaml • Listed the running Pods: kubectl get pods • Deleted the Kubernetes Deployment and its Pods: kubectl delete -f web-deployment.yaml

Implementation:	Step 1: Enable Kubernetes in Docker Desktop Enabled Kubernetes — Enabled Kubernetes in Docker Desktop and verified that the local Kubernetes cluster was successfully initialized. Step 2: Verify kubectl Verified Kubernetes Cluster — Verified that the local Kubernetes cluster was running successfully using kubectl cluster-info and confirmed that the control-plane node was in the Ready state using kubectl get nodes. Step 3: Create the Kubernetes Deployment YAML Created Kubernetes Deployment Manifest — Created the web-deployment.yaml file inside the k8s-manifests directory, defining a Kubernetes Deployment with two replicas of the Flask web application using the Docker image my-flask-app:v2. Step 4: Apply the Deployment Manifest Applied Deployment Manifest — Applied the Kubernetes Deployment manifest using kubectl apply, creating the Deployment resource in the local Kubernetes cluster. Step 5: Verify the Pods Verified Running Pods — Verified that the Kubernetes Deployment successfully created two running Pods, each hosting an instance of the Flask web application. Step 6: Clean Up Deleted Kubernetes Deployment — Removed the Kubernetes Deployment and the Pods it created using the kubectl delete command, cleaning up the local Kubernetes cluster.
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```
1 apiVersion: apps/v1
2 kind: Deployment
3
4 metadata:
5   name: web-app-deployment
6   labels:
7     app: web-app
8
9 spec:
10   replicas: 2
11
12   selector:
13     matchLabels:
14       app: web-app
15
16   template:
17     metadata:
18       labels:
19         app: web-app
20
21     spec:
22       containers:
23       - name: web-app-container
24         image: my-flask-app:v2
25         imagePullPolicy: IfNotPresent
26         ports:
27         - containerPort: 5000
```

```
Error from server (NotFound): error when deleting "web-deployment.yaml": deployments.apps "web-app-deployment" not found
bokhitmahamattom@Bokhits-MacBook-Air k8s-manifests % kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
social-media-app 2/2     2            2           126m
bokhitmahamattom@Bokhits-MacBook-Air k8s-manifests % kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
social-media-app 2/2     2            2           126m
bokhitmahamattom@Bokhits-MacBook-Air k8s-manifests % kubectl version --client
kubectl cluster-info
kubectl get nodes
Client Version: v1.36.1
Kustomize Version: v5.8.1
Kubernetes control plane is running at https://127.0.0.1:56931
CoreDNS is running at https://127.0.0.1:56931/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
NAME          STATUS   ROLES    AGE   VERSION
minikube      Ready   control-plane  138m  v1.35.1
bokhitmahamattom@Bokhits-MacBook-Air k8s-manifests %
```

```
bokhitmahamattom@Bokhits-MacBook-Air k8s-manifests % kubectl apply -f web-deployment.yaml
deployment.apps/web-app-deployment created
bokhitmahamattom@Bokhits-MacBook-Air k8s-manifests % kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
load-generator                      1/1     Running   0           135m
social-media-app-c4ff55b6c-mjzsp    1/1     Running   0           141m
social-media-app-c4ff55b6c-wmsnp    1/1     Running   0           133m
web-app-deployment-6d89664d76-7tf2c 1/1     Running   0           8s
web-app-deployment-6d89664d76-bmgzj 1/1     Running   0           8s
bokhitmahamattom@Bokhits-MacBook-Air k8s-manifests % kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
social-media-app 2/2     2            2           141m
web-app-deployment 2/2     2            2           18s
bokhitmahamattom@Bokhits-MacBook-Air k8s-manifests % kubectl delete -f web-deployment.yaml
deployment.apps "web-app-deployment" deleted from default namespace
bokhitmahamattom@Bokhits-MacBook-Air k8s-manifests %
```

Subject Faculty Name: Prof. Arijit Dutta